

BLOOD TRANSFUSION MANAGEMENT AND HEALTHCARE ANALYTICS

1. Project Overview

This project focuses on cleaning, transforming, and analyzing a **Blood Transfusion Dataset containing 2,000 records** using **Microsoft Excel and Power BI**.

The dataset contains information related to patients, blood groups, hospitals, departments, blood components, transfusion dates, units transfused, hemoglobin levels, treatment outcomes, and transfusion costs.

Excel and Power Query are used for **data cleaning and transformation**, while Power BI is used to perform **data modelling, DAX calculations, analysis, and interactive dashboard development**.

Project Objective

The main objective of this project is to demonstrate end-to-end data analysis by:

- Cleaning and preparing raw healthcare data using Excel and Power Query.
- Analyzing blood transfusion trends and patterns.
- Understanding patient demographics and blood group distribution.
- Identifying departments and hospitals with higher transfusion demand.
- Analyzing blood component utilization.
- Evaluating transfusion outcomes.
- Analyzing healthcare costs related to blood transfusions.
- Creating an interactive Power BI dashboard for effective decision-making.

2. Data Sources

Source Description and Timeline: The dataset used for this project is a **synthetic healthcare dataset**. It contains **2,000 blood transfusion records** covering the period **2024–2026**.

Dataset Name: Blood Transfusion Dataset.xlsx

Domain: Healthcare Analytics / Hospital Management

Data Description: The dataset includes information related to patient demographics, blood groups, hospitals, departments, blood components, transfusion dates, units transfused, hemoglobin levels, treatment outcomes, and transfusion costs.

3. Problem Statement

Blood transfusion management is an important part of healthcare services. Hospitals need to monitor blood demand, blood component utilization, patient conditions, transfusion outcomes, and associated costs.

Therefore, this project aims to analyse blood transfusion data and answer the following business questions:

- Which blood groups have the highest transfusion demand?
- Which departments require the highest number of blood transfusions?
- Which blood components are most frequently used?

- How does the number of transfusions change over time?
- Which hospitals have higher transfusion activity?
- What are the treatment outcomes of blood transfusions?
- How do hemoglobin levels relate to transfusion patterns?
- What is the overall cost of blood transfusion services?

The project provides an interactive Power BI dashboard to support **data-driven healthcare decision-making**.

4. Attribute (Column / Feature) Details

Attribute Name	Data Type	Description
Transfusion_ID	Text	Unique identifier for each transfusion
Patient_ID	Text	Unique identifier for each patient
Age	Whole Number	Age of the patient
Gender	Text	Gender of the patient
Blood_Group	Text	Patient's blood group
Hospital_Name	Text	Name of the hospital
State	Text	State where the hospital is located
Department	Text	Medical department
Transfusion_Date	Date	Date of blood transfusion
Blood_Component	Text	Type of blood component transfused
Units_Transfused	Whole Number	Number of blood units transfused
Donor_Blood_Group	Text	Blood group of the donor
Hemoglobin_Level	Decimal Number	Hemoglobin level of the patient
Outcome	Text	Result or outcome of the transfusion
Cost_INR	Decimal/Currency	Cost of the blood transfusion in INR

5. Tools & Technologies

Microsoft Excel

Excel was used for:

- Identifying missing values
- Checking duplicate records
- Data cleaning
- Data transformation
- Filtering and sorting
- Initial data analysis
- Pivot Tables

Power Query

Power Query was used for:

- Handling missing values
- Changing data types
- Data transformation
- Creating clean and structured data
- Preparing the dataset for Power BI

Power BI

Power BI was used for:

- Data modelling
- Creating calculated columns
- Creating DAX measures
- Data visualization
- KPI development
- Interactive dashboard creation
- Filters and slicers.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed

1. Data Cleaning

The dataset was checked for:

- Missing values
- Duplicate records
- Incorrect data types

- Inconsistent formatting

2. Missing Value Analysis

Raw Data Blood

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	F	G	H	I	J	K	L	M	N	O	P	Q	R
	Hospital Name	State	Department	Insusior	Blood Compone	Units Transfusi	Donor Blood Groi	Hemoglobin Lev	Outcome	Cost If		Unknown	
2	City Care Hospital	Karnataka	ICU	2024-04-12	Red Blood Cells	1	AB+	11.9	Successful	18415			
3	HealthFirst Hospital	Karnataka	ICU	2025-03-11	Red Blood Cells	1	A+	11.3	Successful	20704		16	
4	LifeLine Medical Center	Telangana	Emergenc	2026-05-10	Red Blood Cells	1	B+	6.7	Successful	1666		Age	46
5	City Care Hospital	Andhra Pradesh	Surgery	2024-09-27	Plasma	1	O+	14.6	Successf	22810			
6	LifeLine Medical Center	Kerala	Maternity	2024-03-12	Platelets	1	B+	7.8	Successful	18939		Unit Trans	1
7	LifeLine Medical Center	Tamil Nadu	Oncology	2025-12-01	Red Blood Cells	1	O+	8.4	Mild Reaction	2754		HB	11.3
8	Sunrise Hospital	Karnataka	Medicine	2024-01-21	Platelets	2	B+	12.3	Unknown	5786		Cost	13199
9	Sunrise Hospital	Kerala	ICU	2025-09-06	Red Blood Cells	1	B+	13.2	Successful	9509			
10	Apollo Hospital	Tamil Nadu	Surgery	2024-05-01	Red Blood Cells	1	Unknown	5.2	Successful	5507			
11	Sunrise Hospital	Telangana	ICU	2025-04-11	Red Blood Cells	2	O+	8	Successful	16492			
12	HealthFirst Hospital	Karnataka	Maternity	2024-08-02	Plasma	1	A+	10.4	Successful	9258			
13	LifeLine Medical Center	Karnataka	Emergenc	2024-11-26	Red Blood Cells	1	B+	9.1	Successful	11223			
14	Sunrise Hospital	Kerala	Emergenc	2025-04-03	Plasma	3	O+	16.1	Successful	16090			
15	Apollo Hospital	Andhra Pradesh	Medicine	2024-03-28	Platelets	1	A+	14.8	Successful	17138			
16	Government General Hospital	Telangana	Surgery	2025-01-07	Red Blood Cells	1	B+	12.8	Mild Reaction	13734			
17	HealthFirst Hospital	Andhra Pradesh	Maternity	2024-04-09	Plasma	4	B+	9.2	Successful	15511			
18	Sunrise Hospital	Karnataka	Surgery	2026-05-21	Platelets	1	O+	13.8	Successful	9389			

pivot chart to find gender Raw Data Blood Transfusion Sheet1

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Raw Data Blood

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Conditional Formatting
Easily spot trends and patterns in your data using bars, colors, and icons to visually highlight important values.
Tell me more

	F	G	H	I	J	K	L	M	N	O	P	Q	R
	Hospital Name	State	Department	Insusior	Blood Compone	Units Transfusi	Donor Blood Groi	Hemoglobin Lev	Outcome	Cost If		Unknown	
2	City Care Hospital	Karnataka	ICU	2024-04-12	Red Blood Cells	1	AB+	11.9	Successful	18415			
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4	LifeLine Medical Center	Telangana	Emergenc	2026-05-10	Red Blood Cells	1	B+	6.7	Successful	1666		Age	46
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18	Sunrise Hospital	Karnataka	Surgery	2026-05-21	Platelets	1	O+	13.8	Successful	9389			

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Raw Data Blood

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CommentsShare

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	F	G	H	I	J	K	L	M	N	O	P	Q	R
	Hospital Name	State	Department	Transfusion	Blood Component	Units Transfused	Donor Blood Group	Hemoglobin Level	Outcome	Cost INR		Unknown	
1	City Care Hospital	Karnataka	ICU	2024-04-12	Red Blood Cells	1	AB+	11.9	Successful	18415			
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pivot chart to find gender

Raw Data Blood TransfusionSheet1

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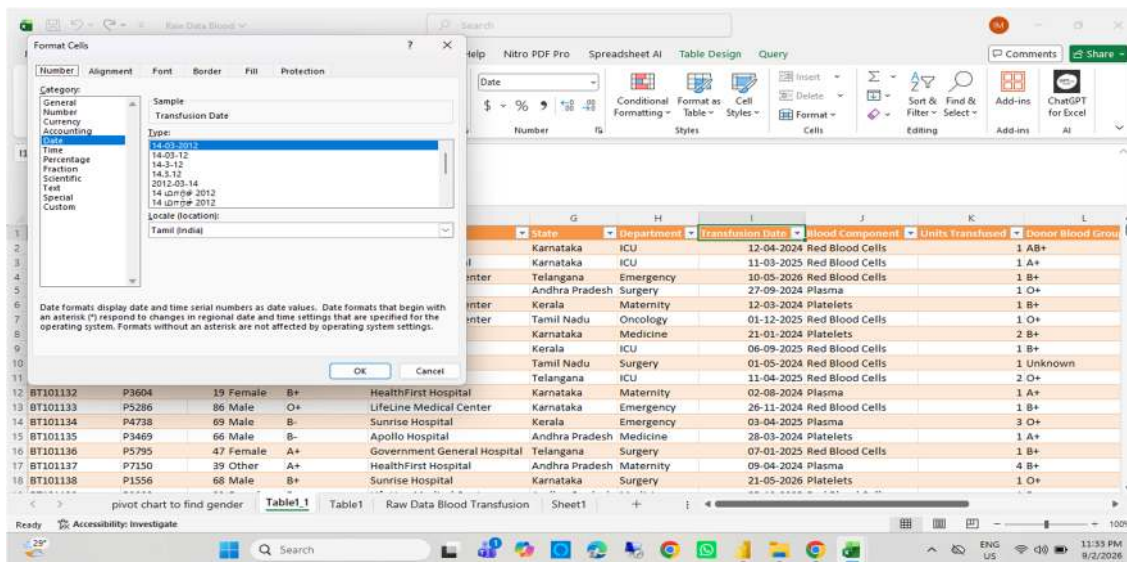
Missing values were handled appropriately during the data cleaning process.

3. Duplicate Check

Duplicate records were checked using the dataset.

The **Transfusion_ID** was treated as a unique identifier.

4. Data Type Conversion



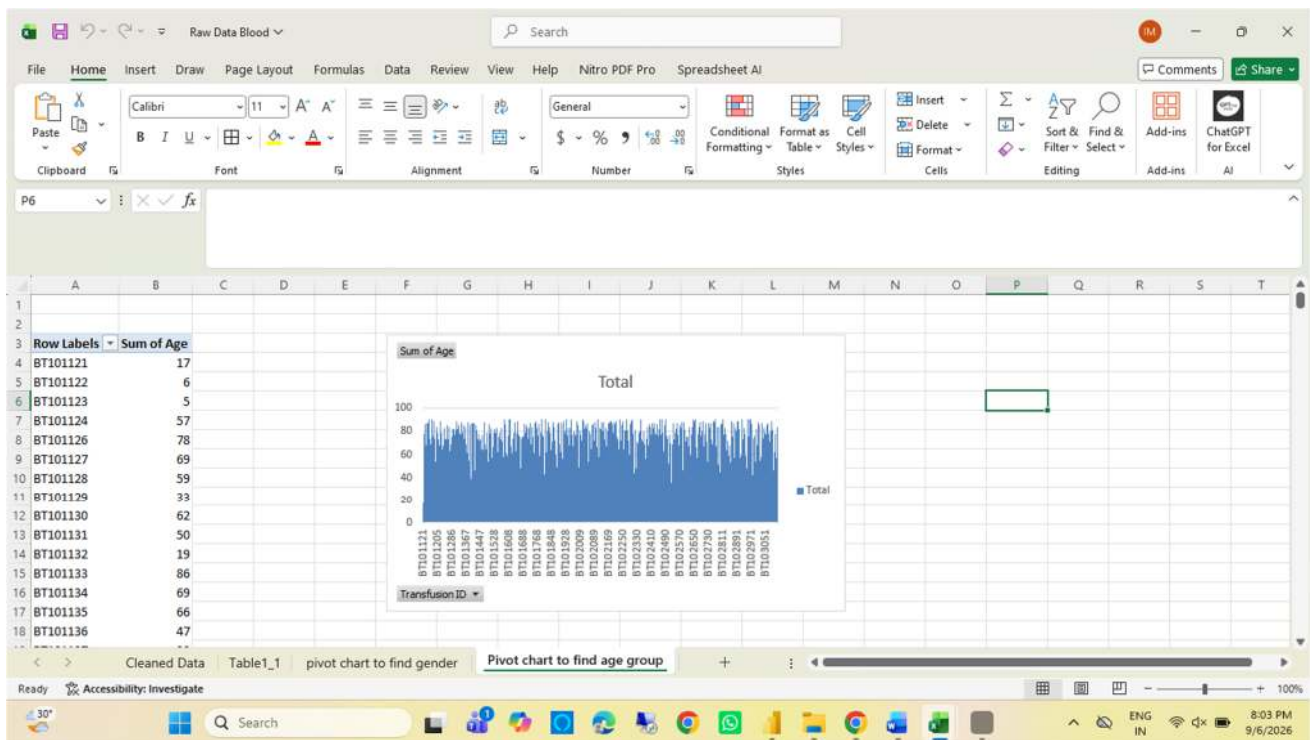
5. Filtering and Sorting

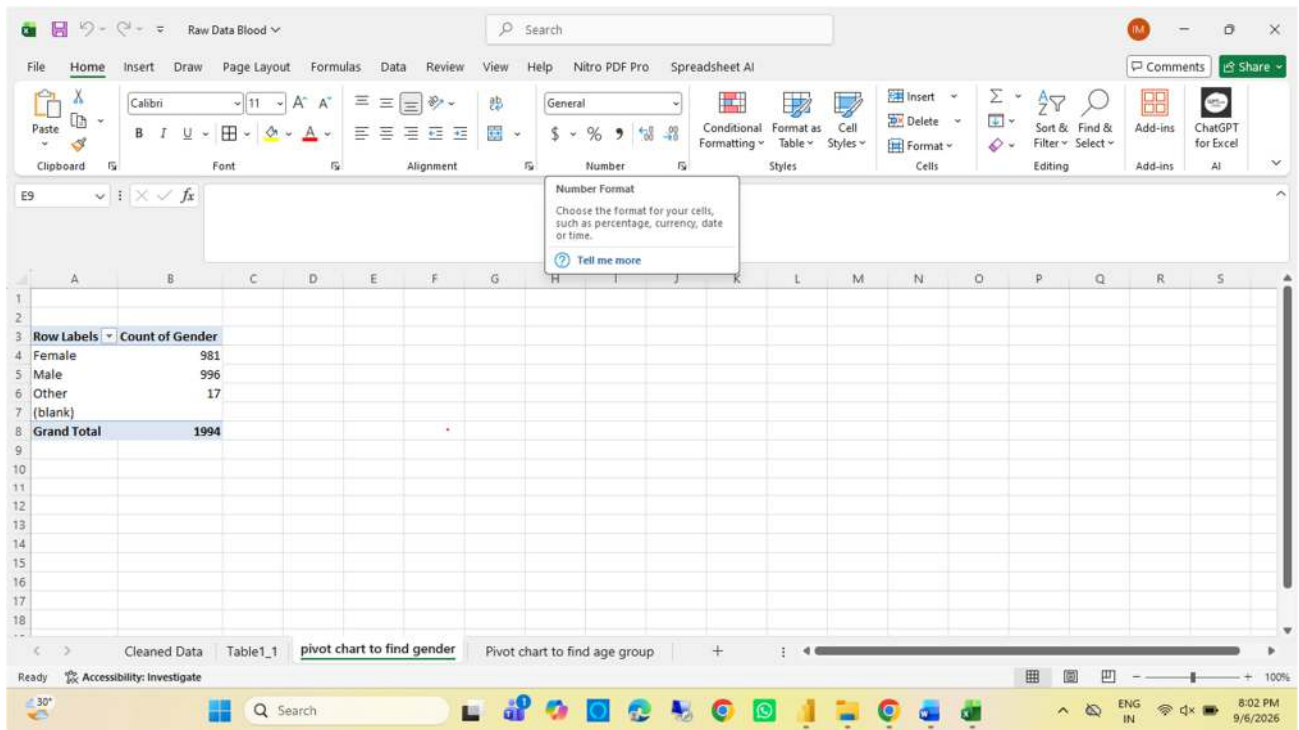
Data was filtered and sorted to analyse:

Hospitals, Departments, Blood groups, Outcomes, Blood components, States

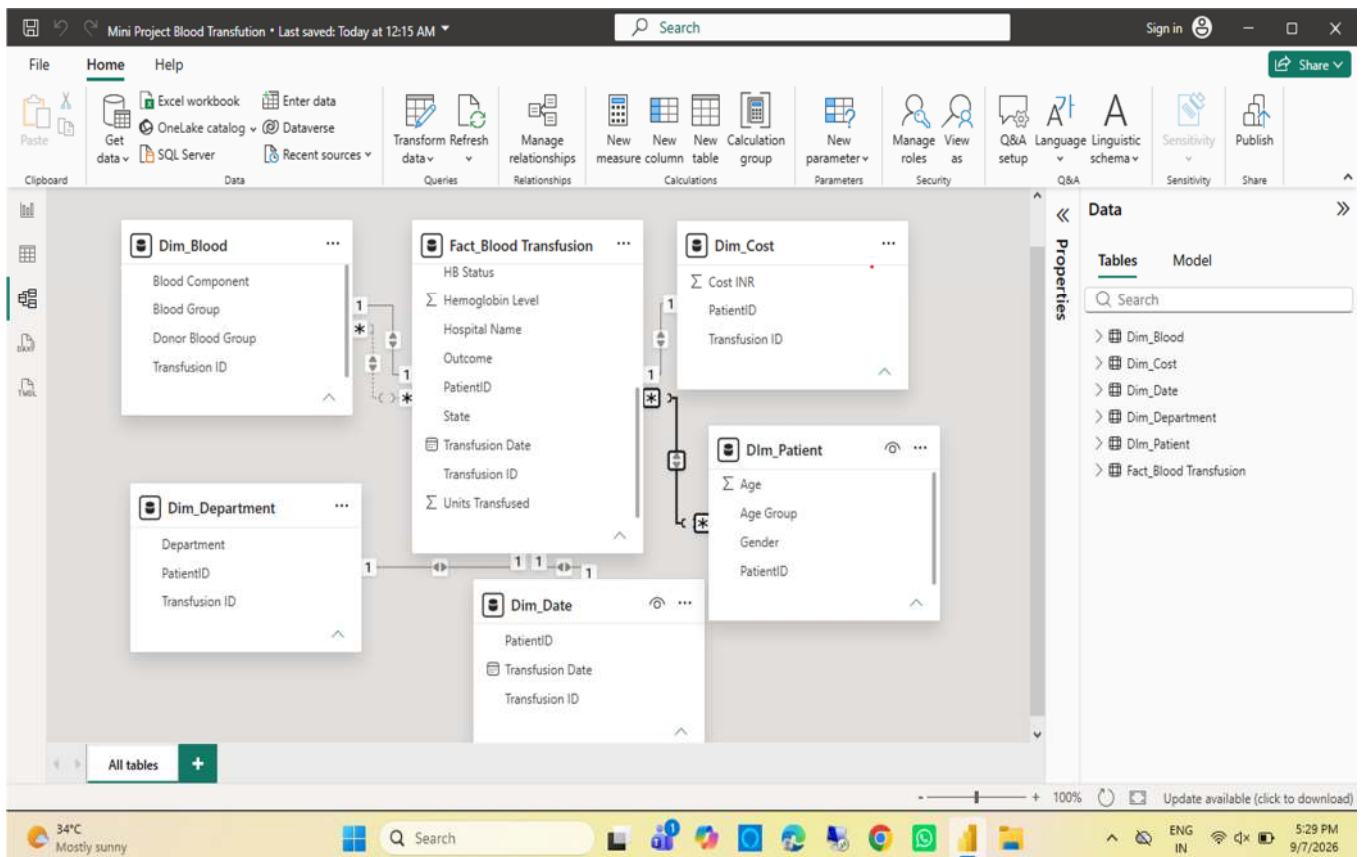
6. Pivot Table Analysis

Pivot Tables can be created in Excel for initial analysis, including:





7. Data Modelling and Dax(PowerBI)



Calculated Columns

1. Age Group

The screenshot shows the Power BI Desktop interface with the 'Column tools' ribbon active. The 'Name' field is set to 'Age Group' and the 'Data type' is 'Text'. The formula bar contains the following DAX formula:

```
Age Group = IF('Raw Data Blood Transfusion'[Age]<18,"Child",IF('Raw Data Blood Transfusion'[Age]<60,"Adult","Senior"))
```

The data table below shows the results of this calculation. The 'Age Group' column has three distinct values: Child, Adult, and Senior.

Department	Transfusion Date	Blood Component	Units Transfused	Donor Blood Group	Hemoglobin Level	Outcome	Cost INR	Age Group
Maternity	Thursday, August 29, 2024	Red Blood Cells	1	B+	8.3	Successful	12382	Senior
Surgery	Thursday, October 9, 2025	Whole Blood	1	B+	6.1	Successful	19435	Child
Maternity	Sunday, April 19, 2026	Red Blood Cells	1	B+	12.2	Successful	8526	Child
Emergency	Monday, April 8, 2024	Red Blood Cells	1	B+	15.3	Successful	8273	Adult
Maternity	Saturday, February 7, 2026	Plasma	1	B+	11.1	Successful	18076	Senior
ICU	Monday, July 28, 2025	Red Blood Cells	1	B+	16.5	Successful	13119	Senior
Maternity	Saturday, October 25, 2025	Platelets	1	B+	5.2	Successful	13420	Senior
Maternity	Friday, October 3, 2025	Plasma	1	B+	5.1	Successful	24423	Adult
ICU	Tuesday, September 3, 2024	Red Blood Cells	1	B+	16.6	Successful	5750	Senior
Emergency	Tuesday, February 17, 2026	Plasma	1	B+	10.8	Successful	15421	Senior
Maternity	Wednesday, October 30, 2024	Red Blood Cells	1	B+	14.9	Successful	4304	Adult
Surgery	Friday, February 23, 2024	Red Blood Cells	1	B+	9.3	Successful	21844	Adult
Surgery	Sunday, June 14, 2026	Red Blood Cells	1	B+	5.4	Successful	2830	Adult
Emergency	Saturday, December 6, 2025	Whole Blood	1	B+	15.4	Successful	20158	Senior
ICU	Thursday, October 3, 2024	Plasma	1	B+	12.8	Successful	13181	Adult
Emergency	Thursday, June 20, 2024	Plasma	1	B+	11.5	Successful	4432	Senior
Emergency	Tuesday, February 6, 2024	Red Blood Cells	1	B+	15.2	Successful	20586	Senior
ICU	Monday, April 8, 2024	Red Blood Cells	1	B+	8.4	Successful	10055	Adult
Oncology	Thursday, April 25, 2024	Whole Blood	1	B+	17.2	Successful	2603	Senior
sh Medicine	Thursday, January 18, 2024	Platelets	1	B+	15.8	Successful	22629	Adult

2. Hemoglobin Status

The screenshot shows the Power BI Desktop interface with the 'Column tools' ribbon active. The 'Name' field is set to 'Hemoglobin Status' and the 'Data type' is 'Text'. The formula bar contains the following DAX formula:

```
Hemoglobin Status = IF('Raw Data Blood Transfusion'[Hemoglobin Level]<8,"Low",IF('Raw Data Blood Transfusion'[Hemoglobin Level]<12,"Normal", "High"))
```

The data table below shows the results of this calculation. The 'Hemoglobin Status' column has three distinct values: Low, Normal, and High.

Transfusion Date	Blood Component	Units Transfused	Donor Blood Group	Hemoglobin Level	Outcome	Cost INR	Age Group	Hemoglobin Status
Thursday, August 29, 2024	Red Blood Cells	1	B+	8.3	Successful	12382	Senior	Normal
Thursday, October 9, 2025	Whole Blood	1	B+	6.1	Successful	19435	Child	Low
Sunday, April 19, 2026	Red Blood Cells	1	B+	12.2	Successful	8526	Child	High
Monday, April 8, 2024	Red Blood Cells	1	B+	15.3	Successful	8273	Adult	High
Saturday, February 7, 2026	Plasma	1	B+	11.1	Successful	18076	Senior	Normal
Monday, July 28, 2025	Red Blood Cells	1	B+	16.5	Successful	13119	Senior	High
Saturday, October 25, 2025	Platelets	1	B+	5.2	Successful	13420	Senior	Low
Friday, October 3, 2025	Plasma	1	B+	5.1	Successful	24423	Adult	Low
Tuesday, September 3, 2024	Red Blood Cells	1	B+	16.6	Successful	5750	Senior	High
Tuesday, February 17, 2026	Plasma	1	B+	10.8	Successful	15421	Senior	Normal
Wednesday, October 30, 2024	Red Blood Cells	1	B+	14.9	Successful	4304	Adult	High
Friday, February 23, 2024	Red Blood Cells	1	B+	9.3	Successful	21844	Adult	Normal
Sunday, June 14, 2026	Red Blood Cells	1	B+	5.4	Successful	2830	Adult	Low
Thursday, December 6, 2025	Whole Blood	1	B+	15.4	Successful	20158	Senior	High
Thursday, October 3, 2024	Plasma	1	B+	12.8	Successful	13181	Adult	High
Thursday, June 20, 2024	Plasma	1	B+	11.5	Successful	4432	Senior	Normal
Tuesday, February 6, 2024	Red Blood Cells	1	B+	15.2	Successful	20586	Senior	High
Monday, April 8, 2024	Red Blood Cells	1	B+	8.4	Successful	10055	Adult	Normal
Thursday, April 25, 2024	Whole Blood	1	B+	17.2	Successful	2603	Senior	High
Thursday, January 18, 2024	Platelets	1	B+	15.8	Successful	22629	Adult	High

3. Cost Category

The screenshot shows the Power BI Desktop interface. The DAX measure bar contains the following formula:

```
1 Cost Category = if('Raw Data Blood Transfusion'[Cost INR]<5000,"Low Cost",if('Raw Data Blood Transfusion'[Cost INR]<15000,"Medium Cost","High Cost"))
```

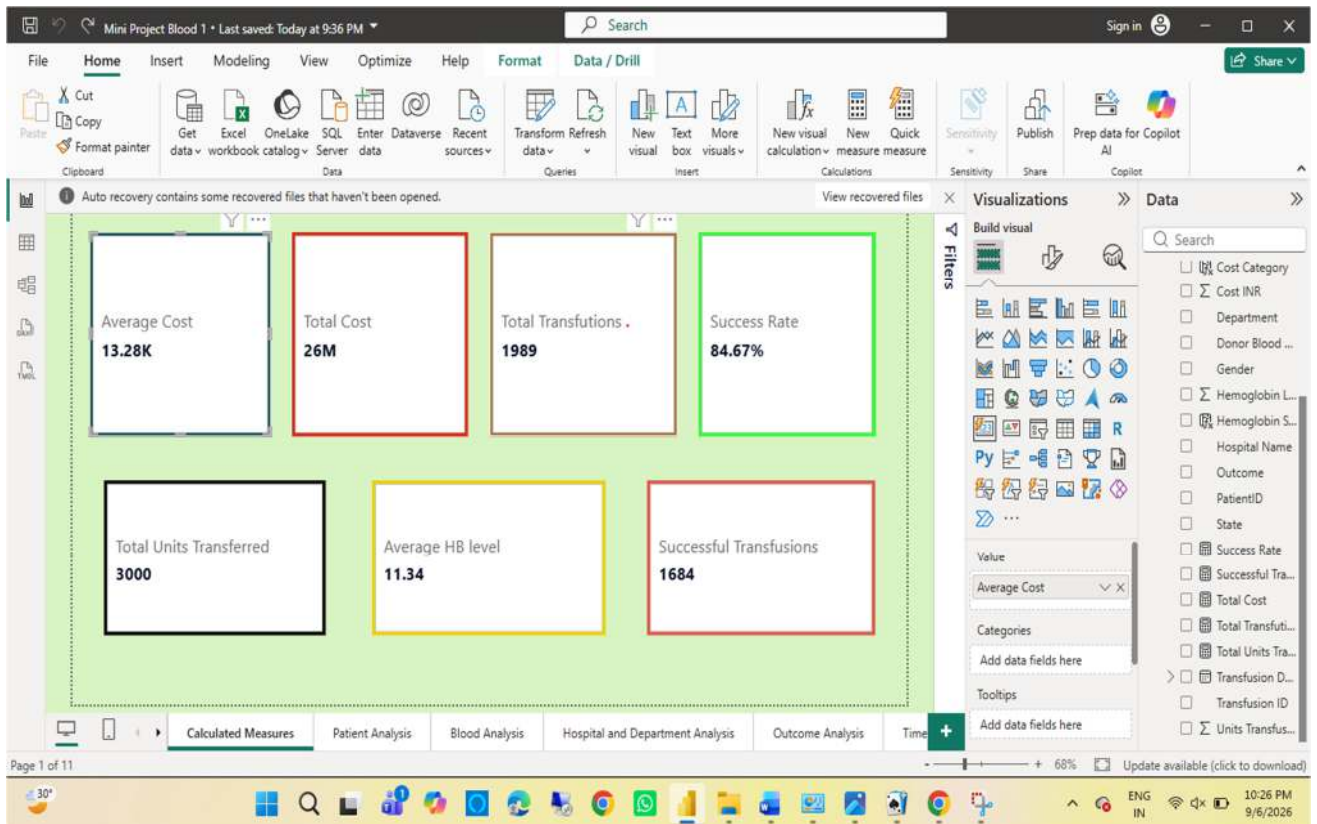
The data table below shows the results of the measure, with columns for various attributes and the calculated Cost Category.

	Blood Component	Units Transfused	Donor Blood Group	Hemoglobin Level	Outcome	Cost INR	Age Group	Hemoglobin Status	Cost Category
2024	Red Blood Cells	1	B+	8.3	Successful	12382	Senior	Normal	Medium Cost
2025	Whole Blood	1	B+	6.1	Successful	19435	Child	Low	High Cost
2026	Red Blood Cells	1	B+	12.2	Successful	8526	Child	High	Medium Cost
2024	Red Blood Cells	1	B+	15.3	Successful	8273	Adult	High	Medium Cost
2026	Plasma	1	B+	11.1	Successful	18076	Senior	Normal	High Cost
2025	Red Blood Cells	1	B+	16.5	Successful	13119	Senior	High	Medium Cost
2025	Platelets	1	B+	5.2	Successful	13420	Senior	Low	Medium Cost
2025	Plasma	1	B+	5.1	Successful	24423	Adult	Low	High Cost
2024	Red Blood Cells	1	B+	16.6	Successful	5750	Senior	High	Medium Cost
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2024	Red Blood Cells	1	B+	14.9	Successful	4304	Adult	High	Low Cost
2024	Red Blood Cells	1	B+	9.3	Successful	21844	Adult	Normal	High Cost
2026	Red Blood Cells	1	B+	5.4	Successful	2830	Adult	Low	Low Cost
2025	Whole Blood	1	B+	15.4	Successful	20158	Senior	High	High Cost
2024	Plasma	1	B+	12.8	Successful	13181	Adult	High	Medium Cost
2024	Plasma	1	B+	11.5	Successful	4432	Senior	Normal	Low Cost
2024	Red Blood Cells	1	B+	15.2	Successful	20586	Senior	High	High Cost
2024	Red Blood Cells	1	B+	8.4	Successful	10055	Adult	Normal	Medium Cost
2024	Whole Blood	1	B+	17.2	Successful	2603	Senior	High	Low Cost
2024	Platelets	1	B+	15.8	Successful	22629	Adult	High	High Cost

Table: Raw Data Blood Transfusion (1,989 rows) Column: Cost Category (3 distinct values)

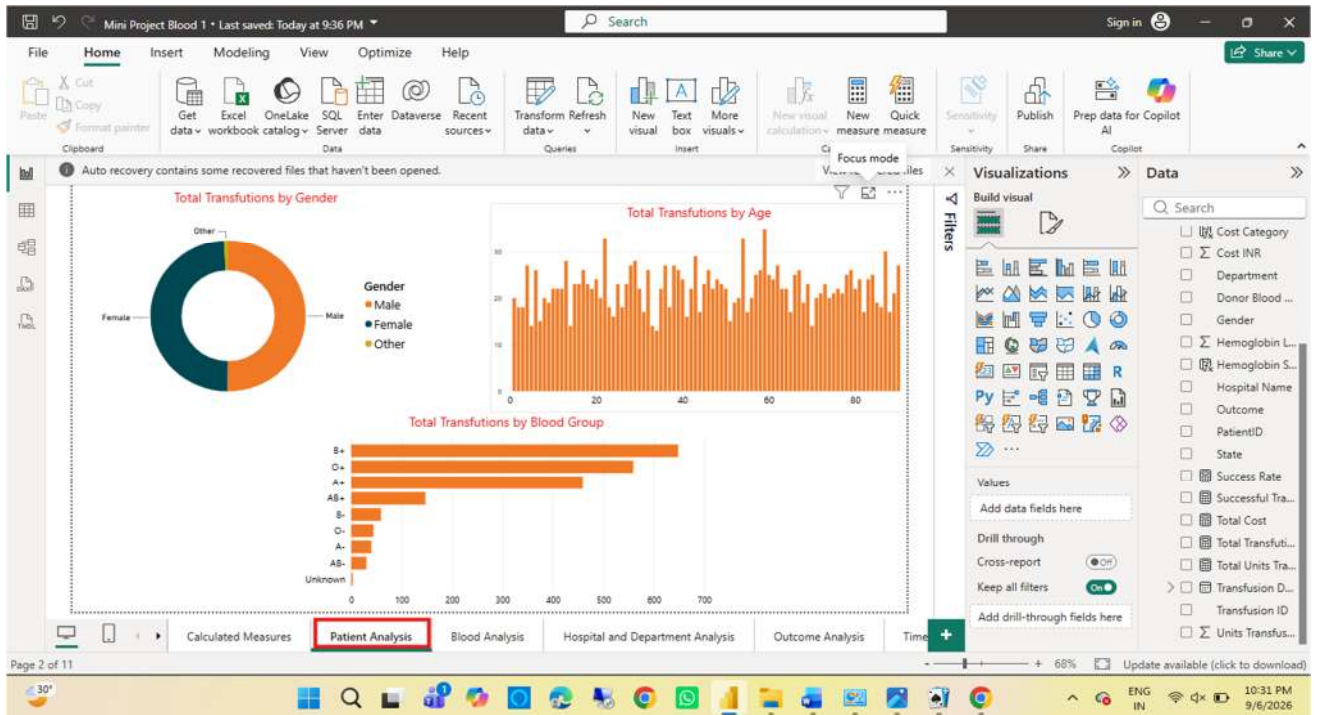
DAX Measures

1. Average Cost = `AVERAGE('Raw Data Blood Transfusion'[Cost INR])`
2. Total Cost = `sum('Raw Data Blood Transfusion'[Cost INR])`
3. Total Transfusions = `COUNT('Raw Data Blood Transfusion'[Transfusion Date])`
4. Success Rate = `DIVIDE('Raw Data Blood Transfusion'[Successful Transfusions],'Raw Data Blood Transfusion'[Total Transfusions],0)`
5. Total Units Transferred = `sum('Raw Data Blood Transfusion'[Units Transfused])`
6. Average HB level = `AVERAGE('Raw Data Blood Transfusion'[Hemoglobin Level])`
7. Successful Transfusions = `CALCULATE(COUNTROWS(FILTER('Raw Data Blood Transfusion','Raw Data Blood Transfusion'[Outcome]="Successful")))`

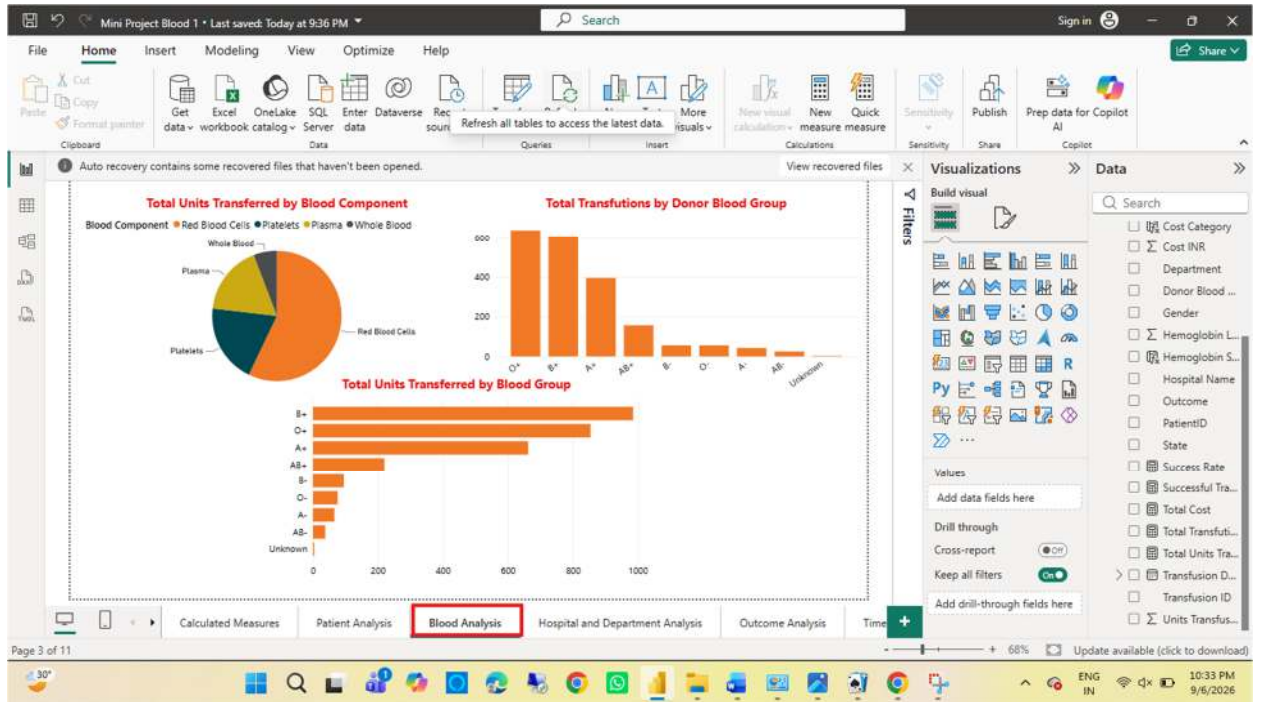


8. Analysis and Visualizations (Power BI)

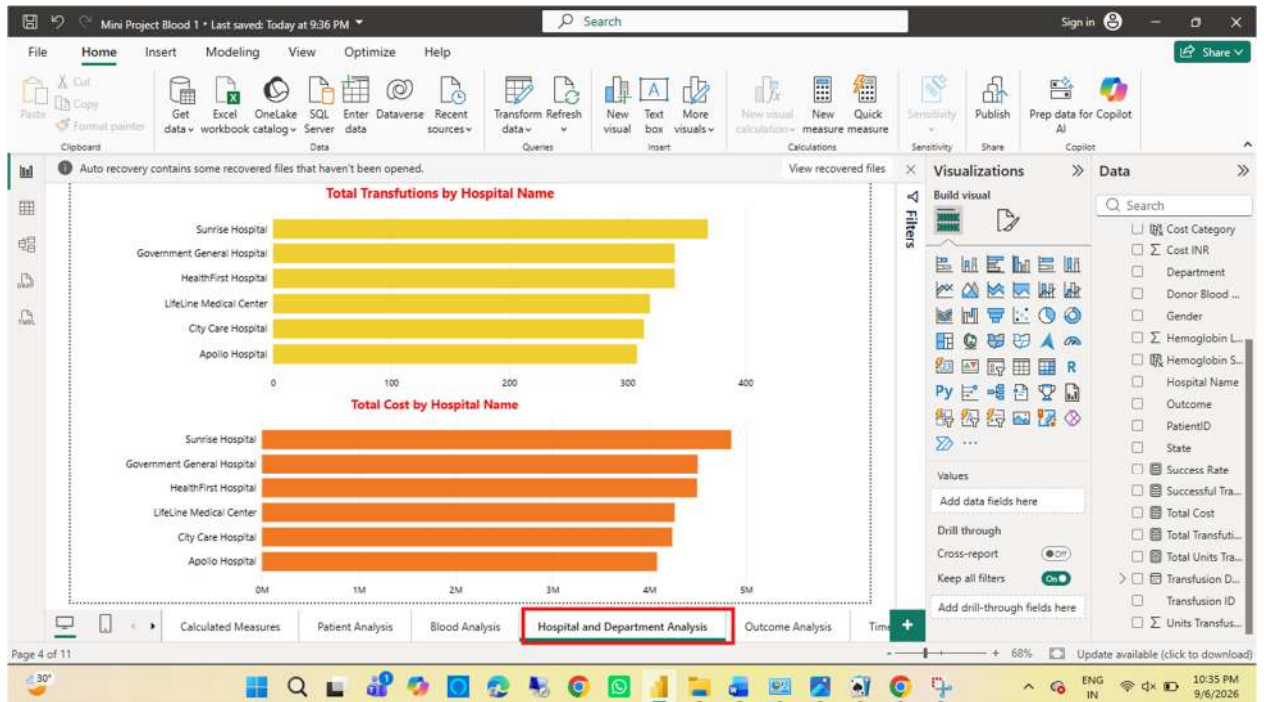
1. Patient Analysis



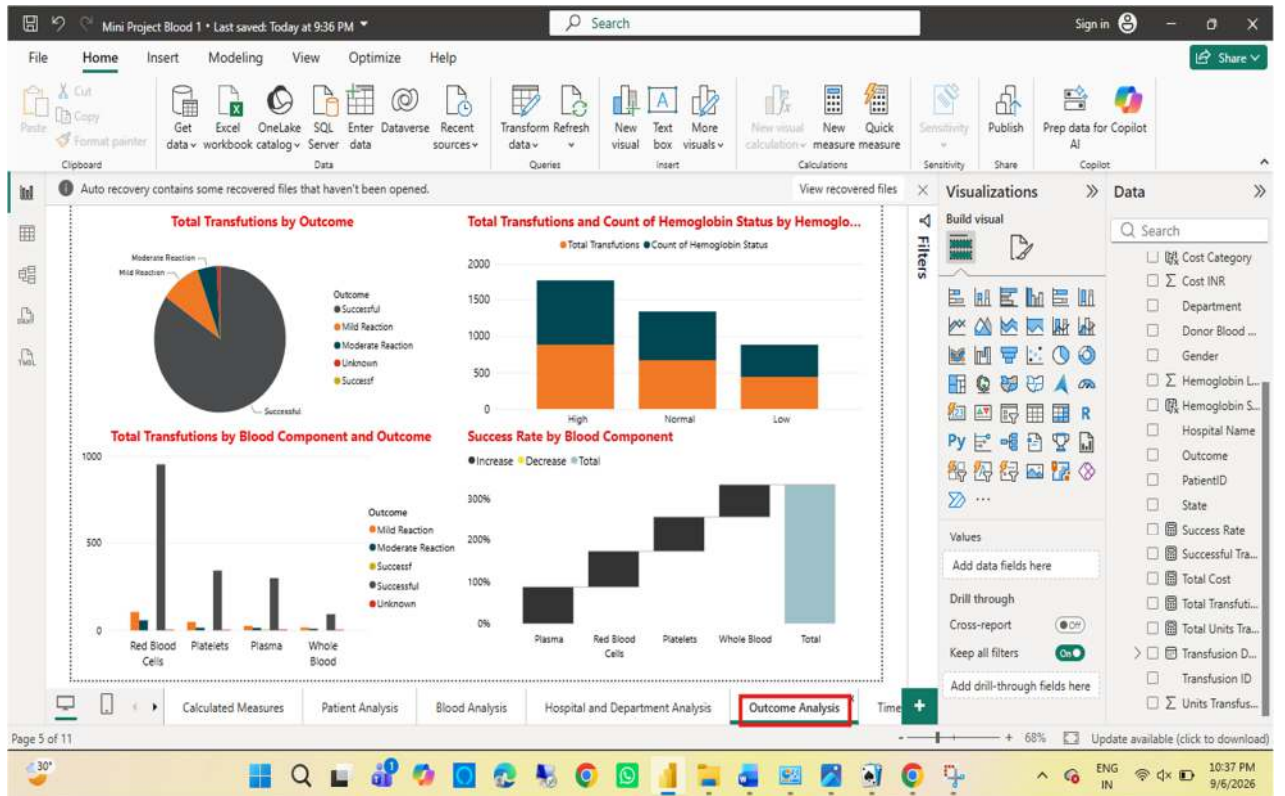
2. Blood Analysis



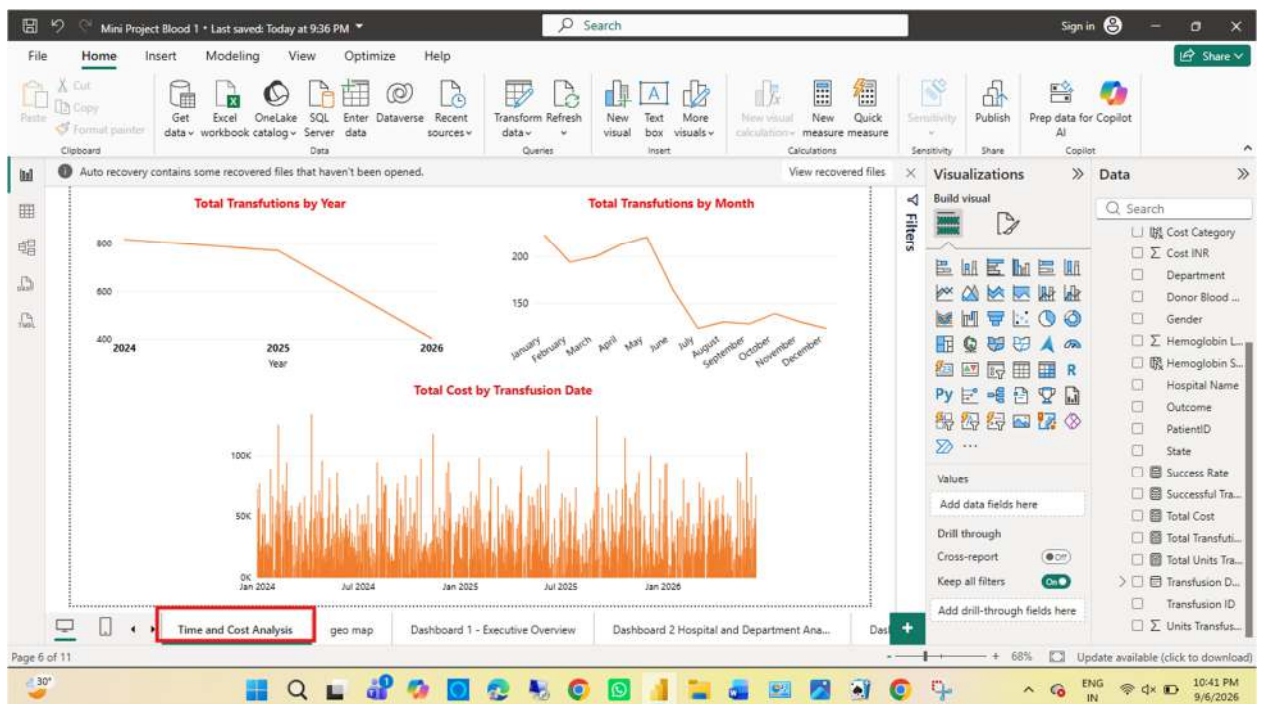
3. Hospital and Department Analysis



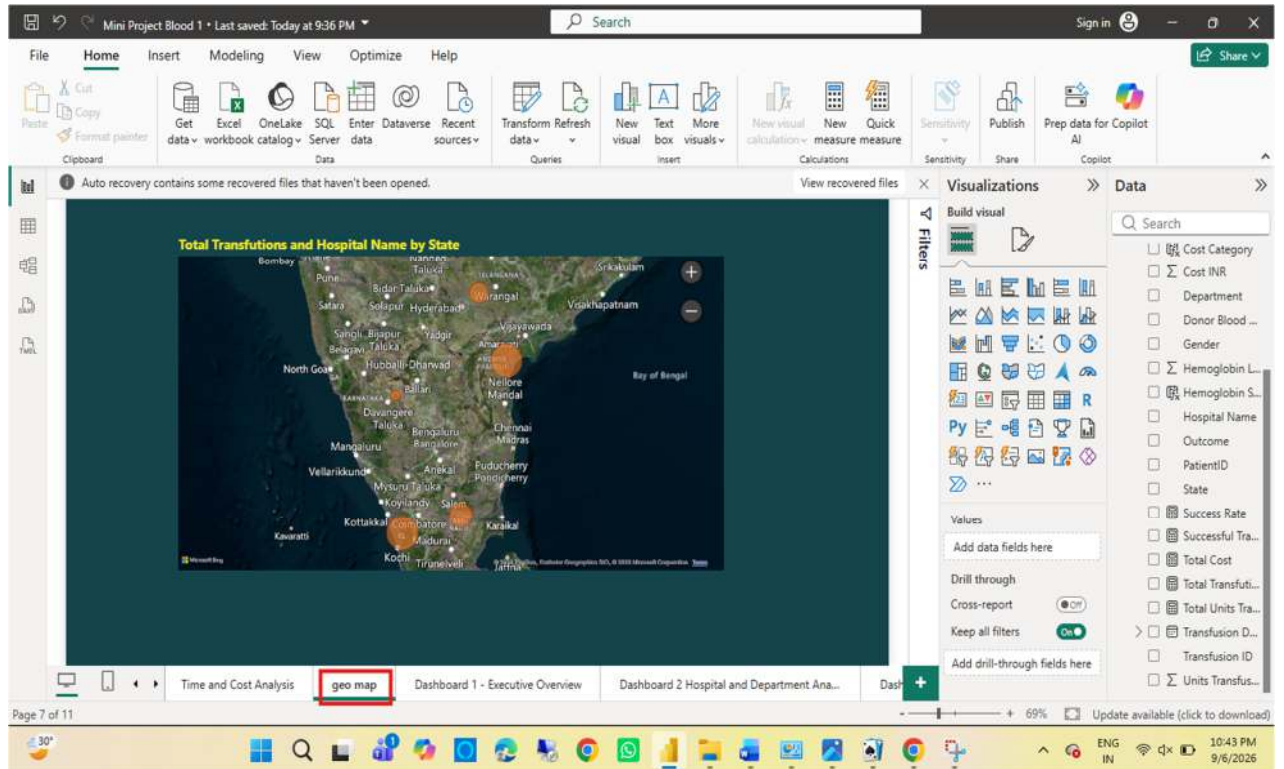
4. Outcome Analysis



5. Time and Cost analysis

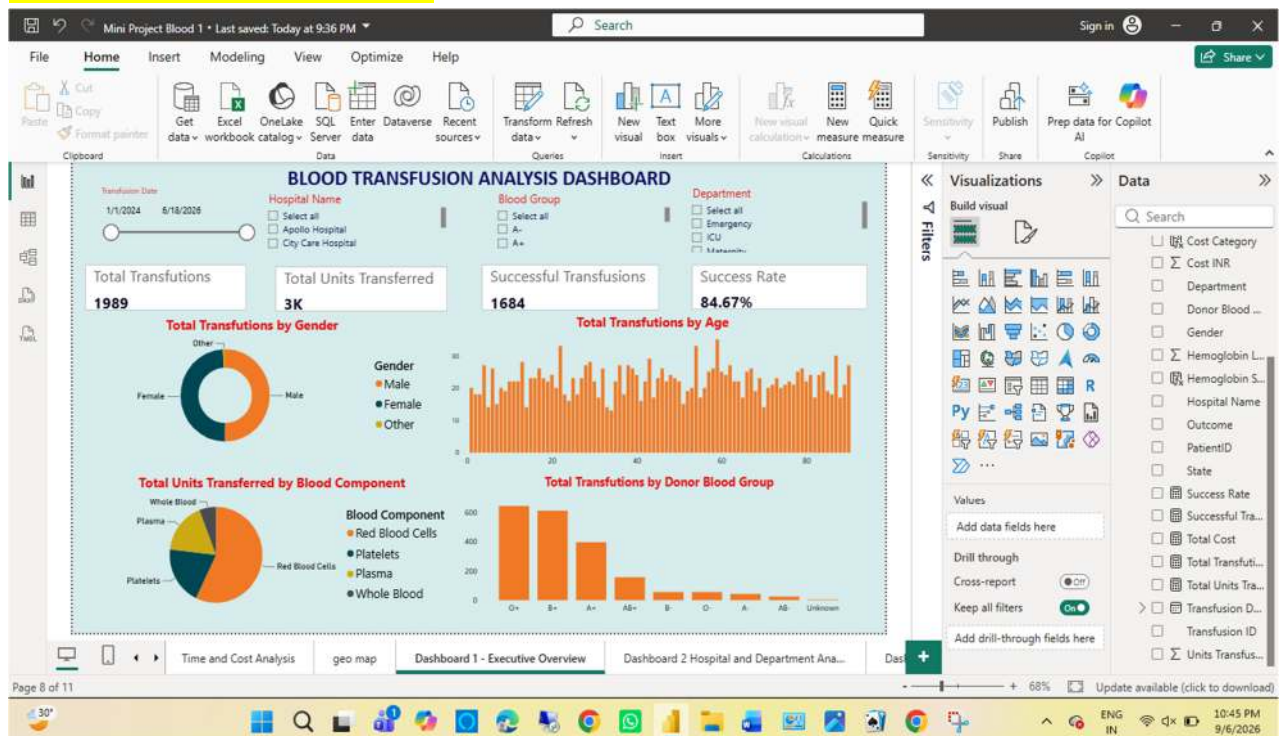


Geo Map

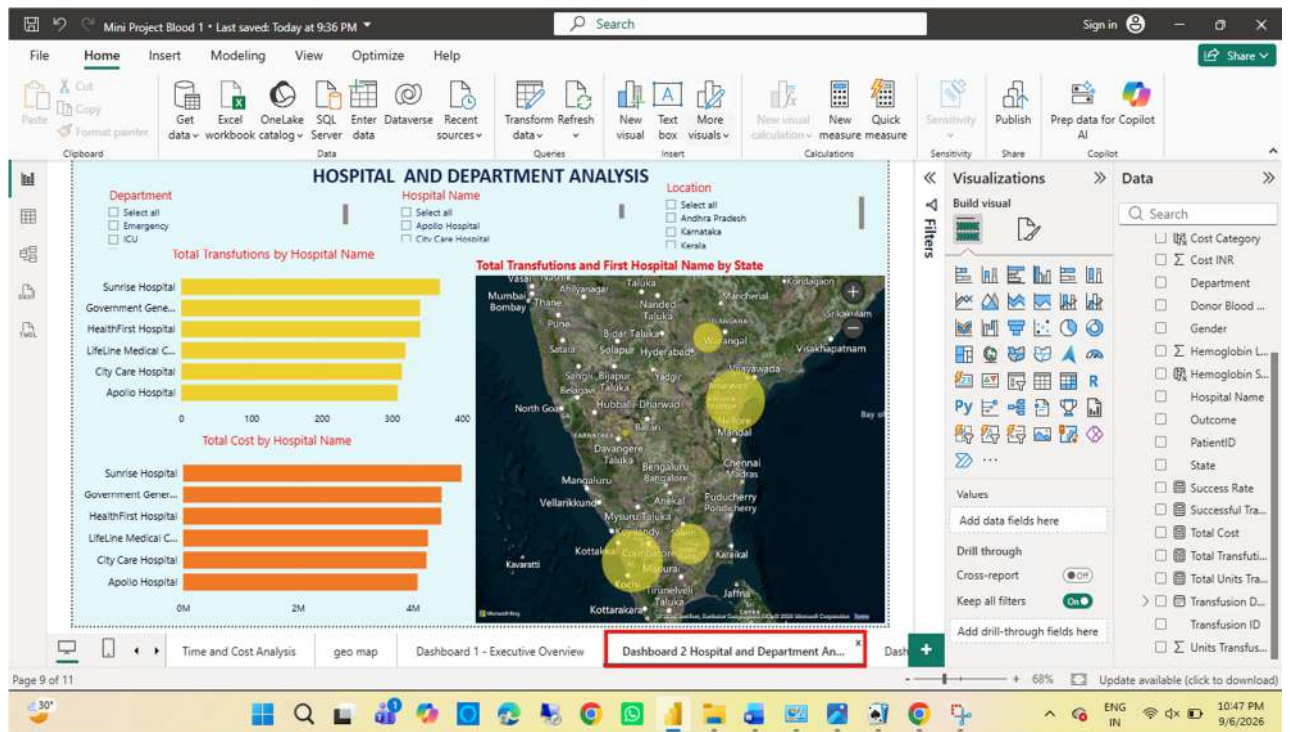


INTERACTIVE DASHBOARD

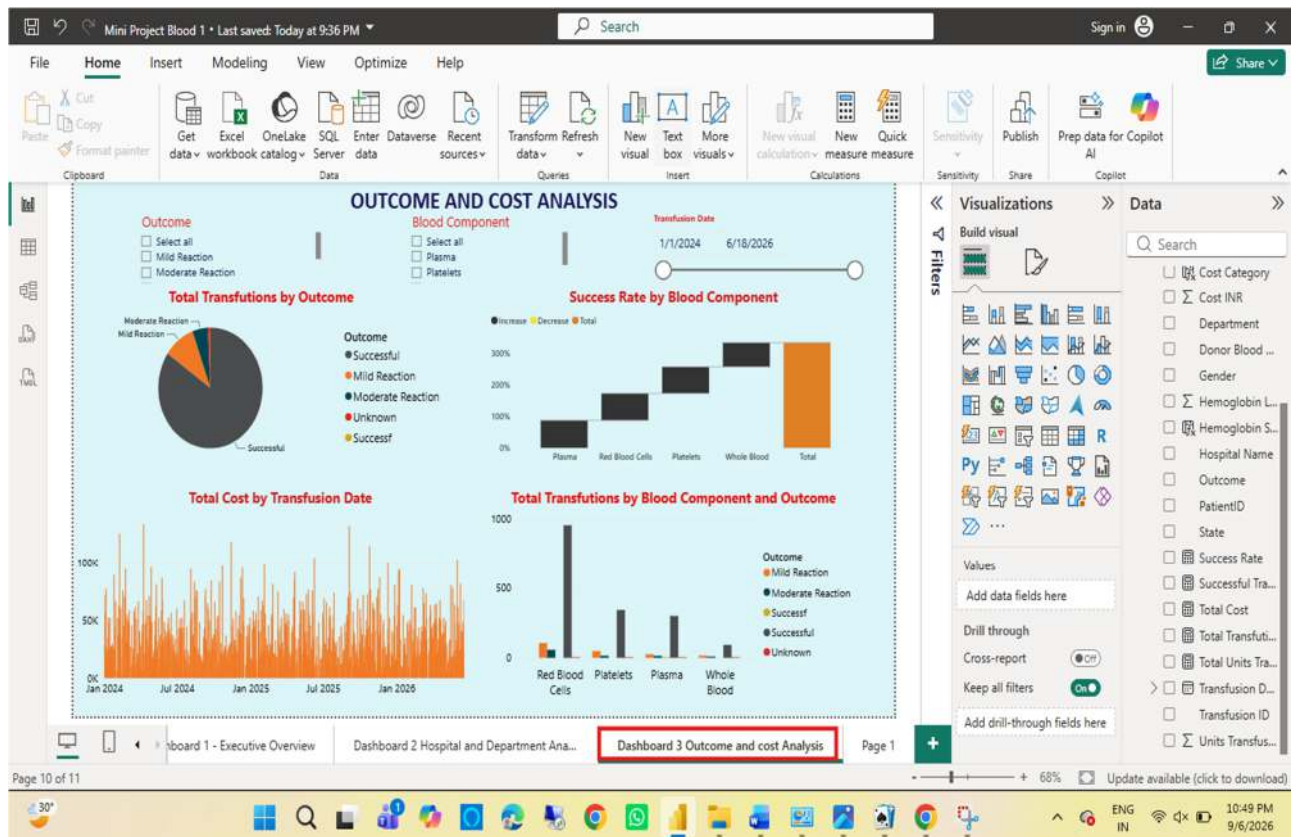
Dashboard 1 Executive Overview



Dashboard 2 Hospital and Departmental analysis



Dashboard 3 Outcome and Cost Analysis



9. Insights and Analysis

Descriptive Analysis

Descriptive analysis focuses on understanding historical data.

Key Dataset Summary

- **Total transfusion records:** 2,000
- **Unique patients:** 1,779
- **Total units transfused:** 3,021
- **Total transfusion cost:** approximately ₹26.48 million
- **Most recorded outcome:** Successful
- The dataset includes records from 2024 to 2026

Outcome Distribution

The majority of transfusions resulted in successful outcomes.

The dataset also includes:

- Mild reactions
- Moderate reactions

This helps healthcare organizations monitor treatment safety.

Diagnostic Analysis

Diagnostic analysis investigates the reasons behind patterns.

Department Analysis

Departments such as:

- ICU
- Emergency
- Surgery

show significant transfusion activity.

Possible reasons include:

- Critical patient conditions
- Emergency medical situations

- Surgical procedures
- Trauma and blood loss

Blood Component Analysis

Different medical conditions require different blood components.

For example:

- Red Blood Cells → Blood loss and anemia
- Plasma → Clotting disorders
- Platelets → Low platelet conditions

Therefore, component utilization patterns can help hospitals plan their blood inventory.

Predictive Analysis

Predictive analysis uses historical patterns to estimate future requirements.

Possible Predictions

Based on historical transfusion trends:

- Hospitals can predict future blood demand.
- Frequently required blood groups can be prioritized.
- Departments with consistently high demand can maintain sufficient blood inventory.
- Seasonal or monthly patterns can help forecast future transfusion requirements.

Prescriptive Analysis

Prescriptive analysis provides recommendations based on insights.

Recommendations

Maintain Adequate Blood Inventory

Hospitals should maintain sufficient stock of frequently required blood groups.

Focus on High-Demand Departments

Departments with higher transfusion requirements should receive priority in blood inventory planning.

Monitor Transfusion Outcomes

Regular monitoring of mild and moderate reactions can improve patient safety.

Control Transfusion Costs

Hospitals should analyse departments with higher costs and identify opportunities for efficient resource management

Final Summary of Key Findings

1.  **B+** has the highest transfusion demand.
2.  **ICU** requires the highest number of transfusions.
3.  **Red Blood Cells** are the most frequently used blood component.
4.  **2024** recorded the highest number of transfusions.
5.  **Sunrise Hospital** has the highest transfusion activity.
6.  Most transfusions resulted in a **successful outcome**.
7.  Hemoglobin levels alone do not show a direct relationship with transfusion frequency.
8.  The total transfusion cost is approximately **₹26.48 million**.

Key Cost Metrics:

Total Cost: ₹26,480,222

Average Cost per Record: ₹13,293

Total Units Transfused: 3,021

10.Conclusion

The **Blood Transfusion Trends and Outcome Analysis** project demonstrates the effective integration of **Microsoft Excel and Power BI for end-to-end healthcare data analysis**. Excel and Power Query were used to clean, transform, and prepare the raw dataset. Power BI was used to build the data model, create calculated columns and DAX measures, and develop an interactive dashboard.

The project provides valuable insights into blood transfusion trends, patient demographics, blood group distribution, department-wise demand, blood component utilization, treatment outcomes, and healthcare costs. The interactive dashboard enables users to explore the data using filters, slicers, drill-down features, and visualizations.